École Polytechnique Fédérale de Lausanne

Analyse III - Lebesgue

MATH-208

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Measure theory

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1.1 Motivation

Studying integration in several variables, the general question we wish to answer is this :

Given some subset $\Omega \in \mathbb{R}^n$, and some real-valued function $f : \Omega \to \mathbb{R}$, is it possible to integrate f on Ω and therefore define $\int_\Omega f$?

In the case $n = 1$, we have already developed the notion of *Riemann integral*, which answers the above question when Ω is an interval $\Omega = [a, b] \subset \mathbb{R}$, and f is Riemann integrable. However, the class of Riemann integral functions is rather unsatisfactorily small and the extension of this notion to higher dimensions is possible but requires quite a bit of effort. For such reasons, we must look beyond the Riemann integral and introduce the notion of the *Lebesgue integral*.

Before we turn to the details, we begin with an informal discussion. In order to understand how to compute an integral $\int_\Omega f$, we must first understand a more basic and fundamental question :

How does one compute the length, area or volume of a subset $E \subset \mathbb{R}^n$?

This question is connected to that integration, because if one integrates the function 1 on the set E, then one should obtain, the length, the area or the volume of E depending on the dimension. To avoid splitting into cases depending on the dimension, we shall refer to the **measure** of E as either.

Ideally, to every subset $E \subset \mathbb{R}^n$ we would like to associate a non-negative number $m(E)$, which will be the measure of E such that,

1. $m(\emptyset) = 0$.
2. $m(E) \in [0, \infty]$.
3. $m(E) \leqslant m(F)$ whenever $E \subseteq F$.
4. $m(E \cup F) = m(E) + m(F)$ if E and F are disjoint.
5. $m(x + E) = m(E)$ for any $x \in \mathbb{R}^n$

Remarkably, it turns our that such a measure does not exist ; one cannot assign a non-negative number to every subset of $\mathbb{R}^n$ which has the above properties. However, we can rescue the situation by only measuring a certain class of sets in $\mathbb{R}^n$, that we will define **measurable set**.

1.2 Outer-measure

Before we construct Lebesgue measure, we first discuss a somewhat naive approach to finding the measure of a set - namely, we try to cover the set by boxes, and then add up the volume of each box. We begin by starting with the notion of an open box.

Définition 1.1 (Open box). An **open box** B in $\mathbb{R}^n$ is any set of the form

$$B = \prod_{i=1}^{n}]a_i, b_i[$$

where $b_i \geqslant a_i$ are real numbers. We define the **volume** $\mathrm{vol}(B)$ of this box to be the number

$$\mathrm{vol}(B) := \prod_{i=1}^{n}(b_i - a_i)$$

Définition 1.2 (Outer measure). If $E \subseteq \mathbb{R}^n$ is a set, we define the **outer measure** $m^*(E)$ of E to be the quantity

$$m^*(E) := \inf\left\{\sum_{j\in J}\mathrm{vol}(B_j) \mid (B_j)_{j\in J} \text{ covers } E; J \text{ countable}\right\}$$

where $(B_j)_{j\in J}$ is a collection of boxes.

Lemme 1.1 (Properties of outer measure)**.** *The outer measure has the following properties :*

1. *The empty set $\emptyset$ has outer measure $m^*(\emptyset) = 0$.*
2. *For every set E we have $m^*(E) \in [0, \infty]$.*
3. *If $A \subseteq B \subseteq \mathbb{R}^n$, then $m^*(A) \leqslant m^*(B)$.*
4. *If E is a subset of $\mathbb{R}^n$, and $x \in \mathbb{R}^n$, then $m^*(x + E) = m^*(E)$.*
5. *If $(A_j)_{j\in J}$ are countable collection of subsets of $\mathbb{R}^n$, then*

$$m^*\left(\bigcup_{j\in J} A_j\right) \leqslant \sum_{j\in J} m^*(A_j).$$

Démonstration :

1. We can cover $\emptyset$ with $]0, \varepsilon[^n$ for any $\varepsilon > 0$. It means

$$m^*(\emptyset) \leqslant \mathrm{vol}(]0, \varepsilon[^n) = \varepsilon^n$$

   We can conclude the $m^*(\emptyset) = 0$ by the fact that ε is arbitrary.
2. Since $\sum_{j\in J}\mathrm{vol}(B_j)$ is non-negative, we know that $m^*(E) \geqslant 0$. However, it quite possible that $m^*(E)$ could equal ∞.
3. Follow from the fact that any cover of B is a cover of A.
4. Similarly, $(B_j)_{j\in J}$ is a cover of E if and only if $(x + B_j)_{j\in J}$ is a cover of $x + E$. Moreover, $\mathrm{vol}(x + B_j) = \mathrm{vol}(B_j)$. We then deduce the claim, because

$$\begin{aligned}
m^*(E) &= \inf\left\{\sum_{j\in J}\mathrm{vol}(B_j) \mid (B_j)_{j\in J} \text{ covers } E\right\} \\
&= \inf\left\{\sum_{j\in J}\mathrm{vol}(x + B_j) \mid (B_j)_{j\in J} \text{ covers } E\right\} \\
&= \inf\left\{\sum_{j\in J}\mathrm{vol}(B'_j) \mid (B'_j)_{j\in J} \text{ covers } x + E\right\} \\
&= m^*(x + E)
\end{aligned}$$

5. For any $j \in J$, by definition of the outer measure as an infimum, it exists $(B_i^j)_{i\in I_j}$ a countable cover of A_j by boxes such that

$$\sum_{i\in I_j}\mathrm{vol}(B_i^j) \leqslant m^*(A_j) + \frac{\varepsilon}{2^j}$$

Taking $(B_i^j)_{i\in I_j, j\in J}$ as a cover of $\bigcup_{j\in J} A_j$, it follows

$$m^*\left(\bigcup_{j\in J}\right) \leqslant \sum_{j\in J}\sum_{i\in I_j} \mathrm{vol}(B_i^j) \leqslant \sum_{j\in J}\left(m^*\left(A_j\right)+\frac{\varepsilon}{2^j}\right) = \varepsilon + \sum_{j\in J} m^*\left(A_j\right)$$

$\square$